MLA0401-Deep Learning for Sustainability

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Experiment 6:

Aim: To demonstrate the performance of knn using wine dataset

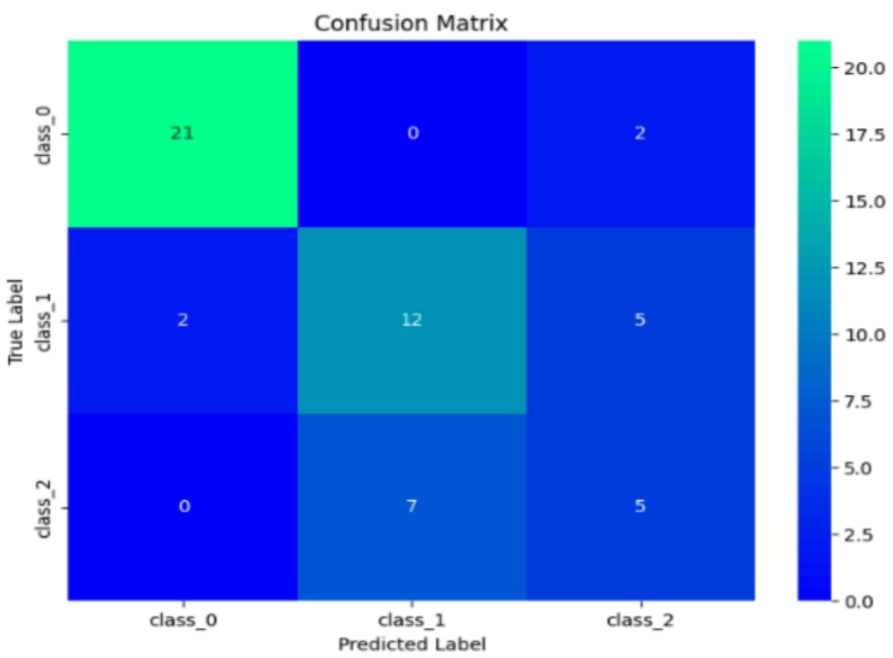
Program:

```
import pandas as pd

import pandas as pd

import matplotlib.pyplot as plt


import seaborn as sns

from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score,confusion_matrix

wine=load_wine()

data=pd.DataFrame(data=wine.data,columns=wine.feature_names)

data['Target']=wine.target

x=data.drop('Target',axis=1)

y=data['Target']

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)

model=KNeighborsClassifier(n_neighbors=5)

model.fit(x_train,y_train)

y_pred=model.predict(x_test)

accuracy=accuracy_score(y_test,y_pred)
```

```
print("accuracy:",accuracy)

conf_matrix = confusion_matrix(y_test,y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='summer',

xticklabels=wine.target_names, yticklabels=wine.target_names)

plt.xlabel('Predicted Label')

plt.ylabel('True Label')

plt.title('Confusion Matrix')

plt.show()
```

Output:

accuracy: 0.7037037037037037



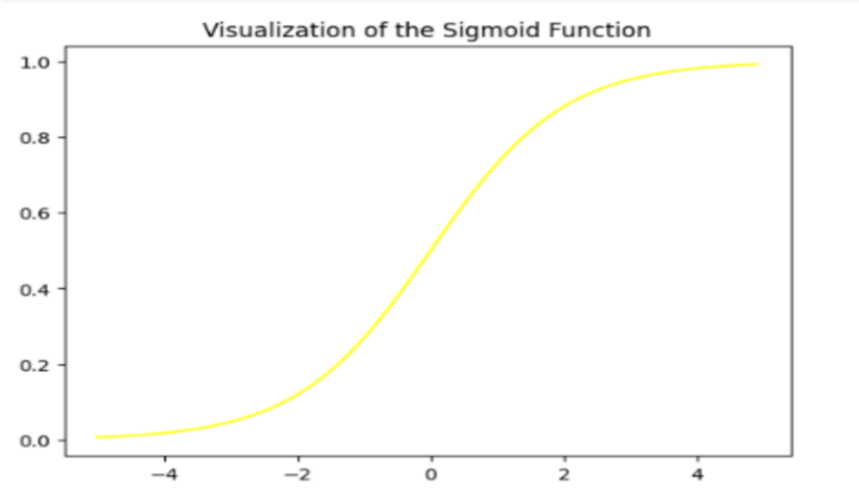
Experiment 7:

Aim: To demonstrate the performance of a logistic regression by using choosen database with python code.

Program:

```
import numpy as np
import matplotlib.pyplot as plt
def sigmoid(z):
return 1 / (1 + np.exp( - z))
plt.plot(np.arange(-5, 5, 0.1), sigmoid(np.arange(-5, 5, 0.1)),color='pink')
plt.title('Visualization of the Sigmoid Function')
plt.show()
```

Output:



Experiment 8:

Aim: To demonstrate the performance of KNN algorithm by using iris dataset

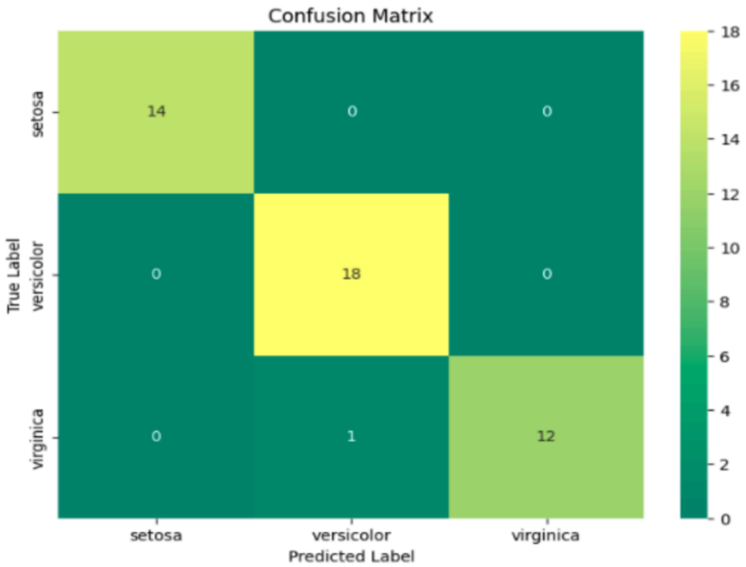
Program:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score,confusion_matrix

iris=load_iris()

data=pd.DataFrame(data=iris.data,columns=iris.feature_names)

data['Species']=iris.target

x=data.drop('Species',axis=1)

y=data['Species']


x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)

model=KNeighborsClassifier(n_neighbors=5)

model.fit(x_train,y_train)

y_pred=model.predict(x_test)

accuracy=accuracy_score(y_test,y_pred)

print("accuracy:",accuracy)

conf_matrix = confusion_matrix(y_test,y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='pink',

xticklabels=iris.target_names, yticklabels=iris.target_names)
```

```
plt.xlabel('Predicted Label')

plt.ylabel('True Label')

plt.title('Confusion Matrix')

plt.show()
```

Output:

accuracy: 0.9777777777777777



Experiment 9:

Aim: To demonstrate the performance of Naïve Bayes algorithm by using iris

dataset

Program:

```
#naive bayes iris
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score,confusion_matrix
iris = load_iris()

data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target
X = data.drop('species', axis=1)
y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred)
```

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='vanimo',

xticklabels=iris.target_names, yticklabels=iris.target_names)

plt.xlabel('Predicted Label')

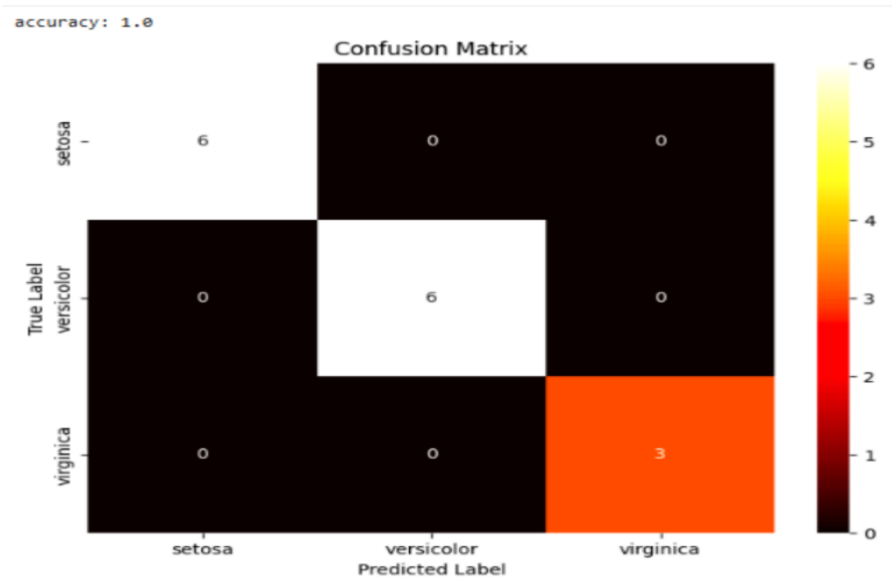
plt.ylabel('True Label')

plt.title('Confusion Matrix')

plt.show()

Output:

accuracy: 1.0



Experiment 10:

Aim: To demonstrate the performance of Logistic Regression using iris dataset

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score,confusion_matrix

iris = load_iris()

data = pd.DataFrame(data=iris.data, columns=iris.feature_names)


data['species'] = iris.target

X = data.drop('species', axis=1)

y = data['species']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,

random_state=1)

model =LogisticRegression(max_iter=200)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("accuracy:",accuracy)

conf_matrix = confusion_matrix(y_test,y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='berlin',
```

xticklabels=iris.target_names, yticklabels=iris.target_names)

plt.xlabel('Predicted Label')

plt.ylabel('True Label')

plt.title('Confusion Matrix')

plt.show()

Output:

accuracy: 0.9777777777777777